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Toward Real-Time Generation With MiniMax H3

8× RTX PRO 5000 · vLLM-Omni

Shunyang Li | Benchmarks measured | 27 Oct 2026



Agenda

01 Toward Real-Time H3 Generation

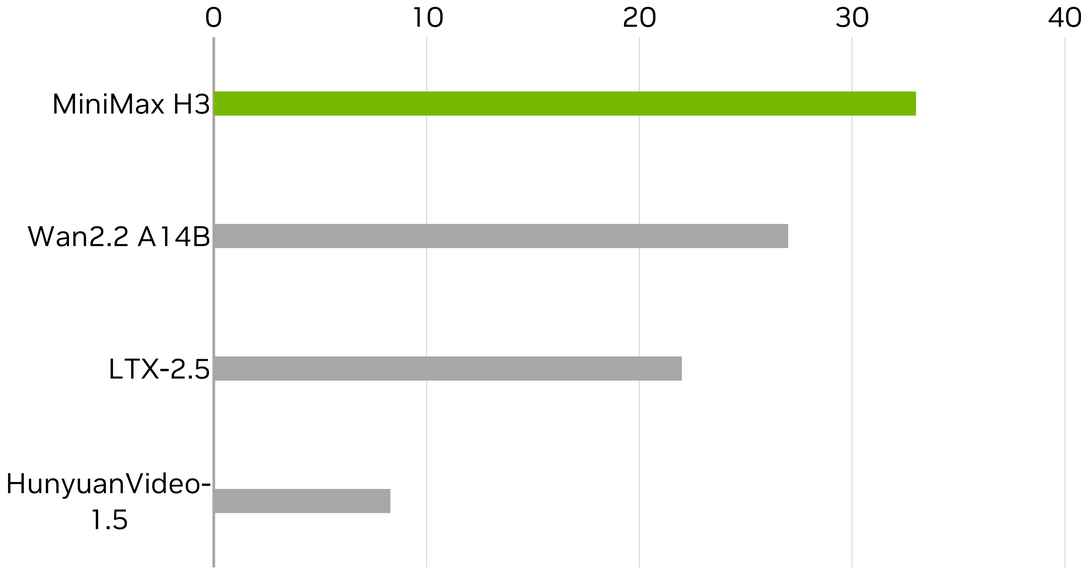
02 vLLM-Omni and ComfyUI Applications



01 · Real-Time H3 Generation

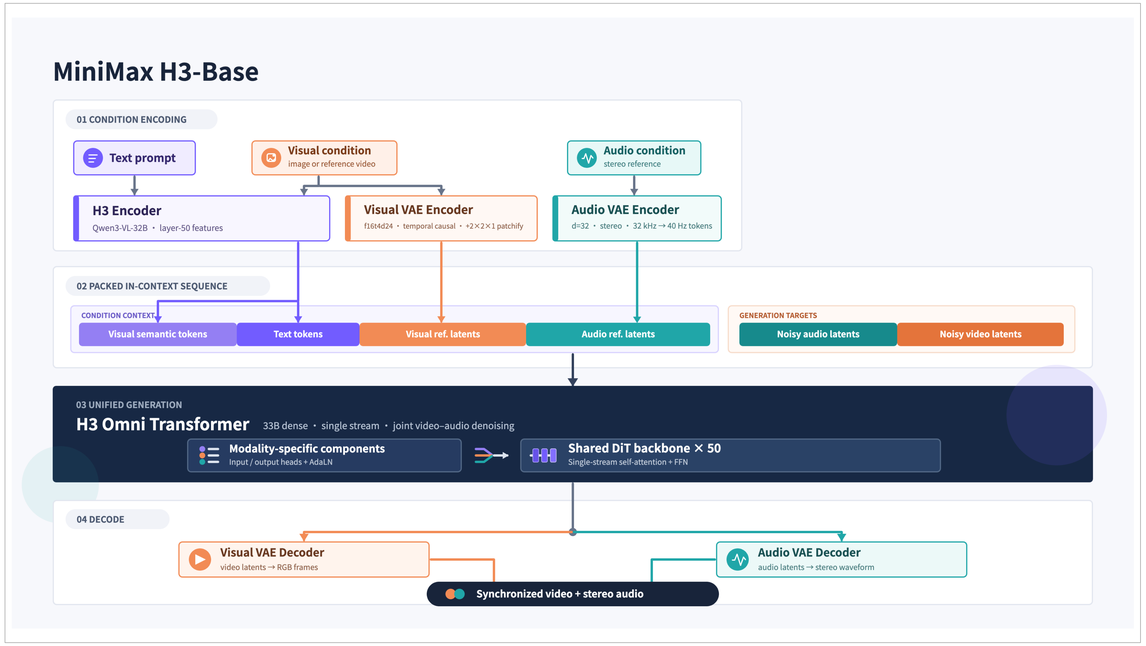
Real-Time Ref2VA Raises the Systems Bar

A 124-frame clip at 24 FPS is real time only when end-to-end latency stays below 5.167 s (RTF < 1)



Generative Backbone · Full Parameters (B)

RTF = latency ÷ media duration; the 124-frame deadline is 5.167 s. H3 has the largest dense backbone in this peer set; Wan2.2 has 27B total and 14B active parameters



Model Capability · Native 768P · 33B Dense

A 32B encoder feeds the 33B Omni Transformer and visual/audio VAEs. Fifty blocks × 50 steps dominate compute; about 13B AdaLN parameters can be cached

Sources: MiniMax H3, Wan2.2, LTX-2.5 and HunyuanVideo-1.5 official releases · author 960×544 Ref2VA workload

SM120 Brings FP4 Compute; PCIe Limits Scale

All discrete SM120 cards pair fifth-generation Tensor Cores with GDDR7 and PCIe Gen5; none provides GPU-to-GPU NVLink

SKU	Compute	Memory	GPU I/O / Fabric
RTX PRO 5000 ×8 Blackwell	SM120 · 110 SMs 5th-gen Tensor · FP4	72 GB GDDR7 ECC 48 GB variant	PCIe Gen5 ×16 Planned: RDMA via NIC*
RTX PRO 6000 WS / Server	SM120 · 188 SMs 5th-gen Tensor · FP4	96 GB GDDR7 ECC	PCIe Gen5 ×16 No GPU NVLink
GeForce RTX 5090	SM120 · 170 SMs 5th-gen Tensor · FP4	32 GB GDDR7	PCIe Gen5 ×16 No NVLink
DGX Spark · GB10	SM121 · 48 SMs 5th-gen Tensor · FP4	128 GB unified LPDDR5X	NVLink-C2C in SoC 200 Gb/s ConnectX-7

* Planned system path; GPUDirect RDMA requires a compatible NIC and PCIe topology · Sources: NVIDIA product specs, CUDA GPU table, GPUDirect RDMA guide

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Current Configuration: Latency Breakdown

Final slide 18 group · standalone Ref2VA · initialization and warmup excluded

OUTPUT	VIDEO PER REQUEST	PARALLELISM	HARDWARE
960 × 544	5.167 seconds	8 GPUs	SM120 / PCIe
Reference image: 1344 × 768	124 frames at 24 FPS	Ulysses 8 · Encoder TP8 · VAE tiles 8	PRO 5000 reference node · RDMA / IB

5.418 s
Unprofiled HTTP median · n = 3

5.664 s
One warmed Nsight request

Profiled request: rank 0 stage envelopes

Text + vision encoder	0.2485 s
Reference image + audio encoder	0.1303 s
DiT · eight evaluations	4.3024 s
Decode + MP4	0.8481 s
Other / request-boundary residual	0.1343 s

CPU NVTX ranges include waits; values are rounded.
Nested VAE decoder: 0.572 s, included in Decode + MP4.

What is enabled

Model
QAD NVFP4 DiT · trained VSA · native FP4 attention

Communication
RDMA direct layout · V / gate overlap · 2 / 3 / 2 heads

Execution
Head compute CUDA Graph · 1,200 replays / rank

Decoder
36L NVFP4 VAE · 216 linears · BF16 autocast · MP4 pre-encode

AdaLN
256 MiB exact cache · 366 hits / 42 misses · weights resident

Remaining GPU work in DiT · GPU 0

Pack / unpack / O-bundle: 158.9 ms
Gate assembly: 32.7 ms · D2D copies: 32.2 ms
All H2D copies: 0.36 ms / 9.03 MB; no large AdaLN weight staging.

GPU activity is already inside DiT; it is not additive E2E time. Sub-5 s target is not met. Hardware label follows project records.

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Five-Second Avatar: Show the Result

Demo focus · reference identity, expression, native speech and continuity

01 · Natural presence

Start from the reference portrait.
Show a short reading / idle segment.

02 · Speech and expression

Play a short spoken response with audio.
Inspect face motion and lip synchronization.

03 · Continuous generation

Continue the same scene across segment boundaries.
Show generation and playback on one timeline.

Historical avatar: 960 × 544 · 24 FPS · 4 steps · 8 GPUs · BF16 RDMA

119 new frames / 4.958 s per continuation · 3.706 s median application-ready time, n = 13.
Live rerun and playback validation pending. These timings belong to the separate four-step avatar recipe.

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Three Ideas Behind Faster Generation

Technical overview · detailed algorithms and source excerpts remain available on hidden slides

Less model work

Few-step distillation reduces evaluations.
Sparse attention concentrates work on selected regions.
Quantized inference lowers arithmetic and memory cost.

Overlap across eight GPUs

Ulysses distributes attention across heads.
RDMA carries the exchanges.
Head bands overlap transfer with attention compute.

Keep the output path moving

CUDA Graph replays stable head-band compute.
Exact AdaLN caching avoids repeated projections.
NVFP4 VAE and MP4 pre-encoding reduce output cost.

These are optimization families. Use each demo’s own recipe and measurements; the four-step avatar and slide 18 are distinct workloads.

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Optimization Roadmap: 5 s Live Target

One fixed Ref2VA case · Eight cumulative groups · Seven measured benchmarks

960 × 544 · 124 frames / 24 FPS · 8 GPUs · seed 43

Model Compute		E2E (s)
0	**Historical Base50** BF16 DiT · dense · 50 steps Default offload · local NCCL	247.956
1	**PDD: 50 to 8 Steps** Reduce denoiser evaluations Eight-step Ref2VA schedule	42.962
2	**VSA** Select fine-attention blocks Keep global coarse context	26.199
3	**QAD + Native FP4 Attention** NVFP4 DiT · unified prefix / video Device-aware SM scheduling	19.351

Communication and Output		E2E (s)
4	**Ulysses Communication** RDMA + direct layout + V / gate overlap IB for remaining NCCL collectives	6.831
5	**2 / 3 / 2 Head Pipeline** Overlap communication and attention Restore heads before output projection	6.472
6	**CUDA Graph** Replay each head band's compute Reduce repeated CPU launch overhead	6.201
7	**NVFP4 36L VAE + MP4** Native 36-layer decoder to NVFP4 MP4 pre-encoding pipeline	5.418

Target: full request < 5.000 s · Not reached

Seven groups measured · 3 formal requests each · Initialization and warmup excluded

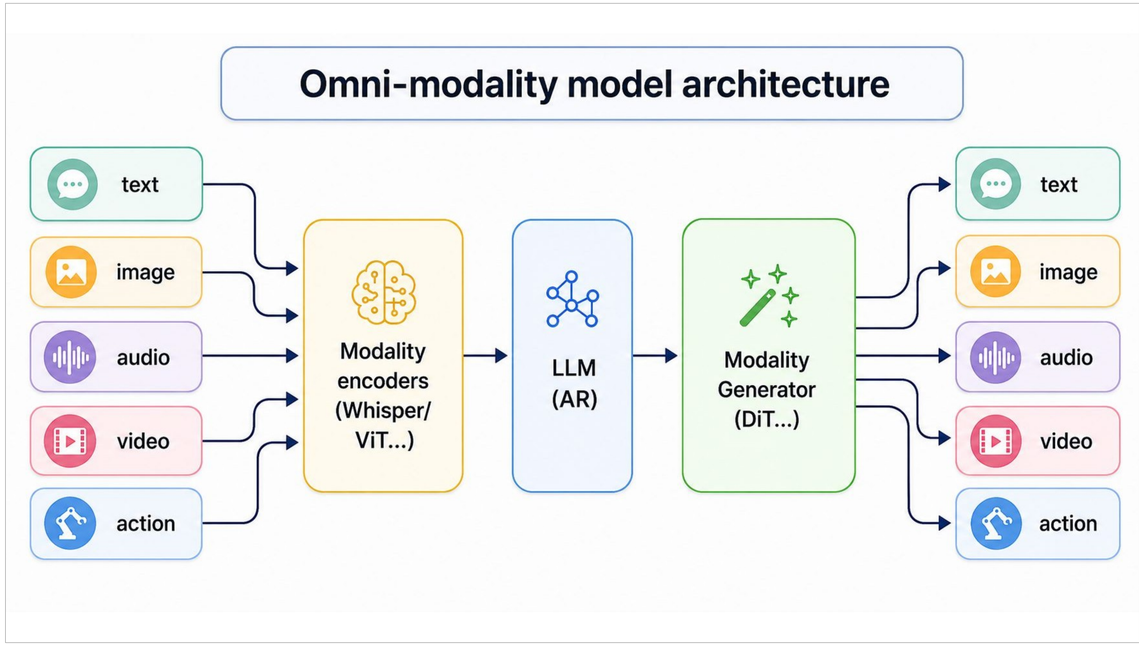
Base50 default offload retained · optimized BF16 baseline pending · warm request to complete MP4

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02 · ComfyUI Apps With vLLM-Omni

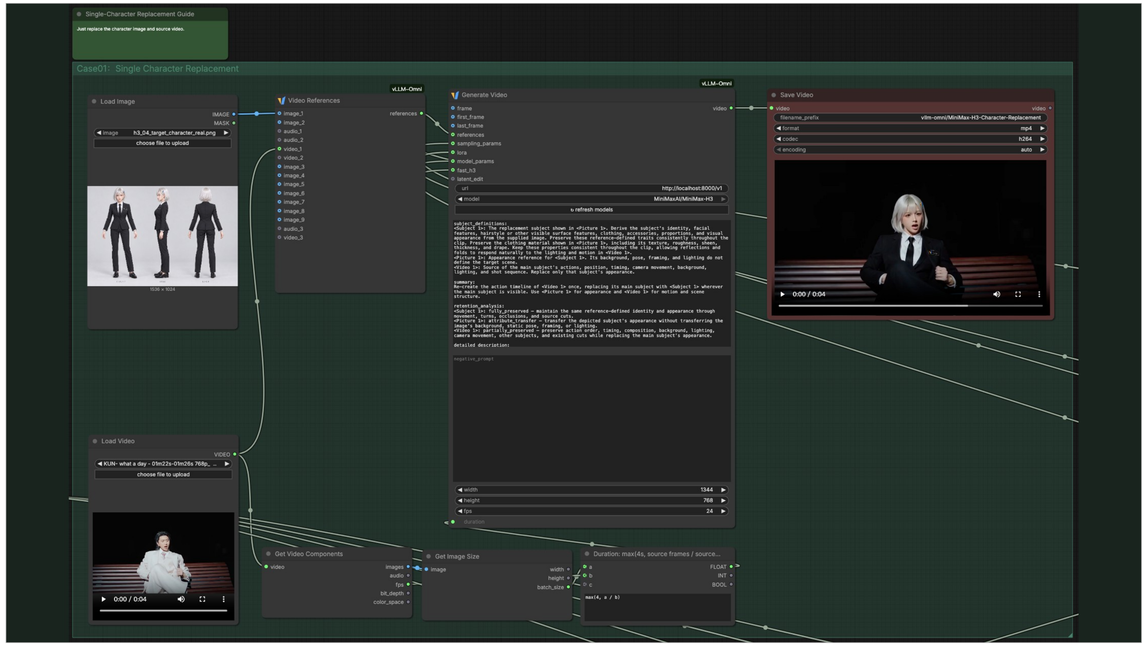
vLLM-Omni Brings Multi-GPU H3 to ComfyUI

PR #9194 adds character replacement, continuation-aware references, and chainable LoRA controls



Official vLLM-Omni Architecture

ComfyUI authors the workflow; vLLM-Omni runs scheduling, model execution, and multi-GPU deployment



ComfyUI Workflow · PR #9194

The workflow supports single- and multi-character replacement, source-derived duration, long-video continuation, and LoRA controls

Sources: vLLM-Omni v0.26 public deck, slide 22 · vLLM-Omni PR #9194

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